Model Predictive Control for Pendulum Swing-Up

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Abstract

We study model-based control on the pendulum swing-up task using Model Predictive Control (MPC) with a known dynamics model [1–3]. At each time step, the controller samples candidate action sequences, predicts future trajectories, and selects the sequence with the lowest predicted cost. We investigate how planning horizon and control penalty affect swing-up performance, trajectory quality, and control effort. Our results show that MPC can solve the task reliably when the planning horizon is sufficiently long, while stronger control penalties reduce actuation but may hinder successful swing-up.

1 Introduction

Model-based control uses an explicit model of system dynamics to select actions that optimize future behavior. A standard example is Model Predictive Control (MPC), where the controller repeatedly plans over a finite horizon, executes the first action of the best plan, and replans at the next time step [1–3]. This makes MPC especially attractive for nonlinear control tasks, where future consequences of actions matter [1,2].

In this project, we study MPC on the pendulum swing-up task. This task is a useful benchmark because it requires non-myopic control: to reach the upright position, the controller often needs to first move the pendulum away from the target in order to build momentum. As a result, short-sighted control strategies can fail even though the system itself is simple. More broadly, such sequential decision-making problems are closely related to ideas from optimal control and dynamic programming [4,6,7].

We use a known pendulum dynamics model and implement MPC with random shooting. At every decision step, the controller samples many candidate action sequences, rolls them out under the dynamics model, evaluates their trajectory cost, and executes the first action of the best sequence. This approach is simple to implement, easy to interpret, and sufficient for a low-dimensional control problem such as pendulum swing-up. Sampling-based MPC of this form is also closely related to information-theoretic and path-integral control methods studied in prior work [8,9].

The main research question is:

> How do planning horizon and control penalty affect the performance of MPC on the pendulum swing-up task?

We evaluate performance in terms of swing-up success, final angle error, control effort, and cumulative cost. We expect that longer planning horizons will improve swing-up success by allowing the controller to reason farther ahead [1–4], while larger control penalties will reduce aggressive actuation but may make swing-up more difficult [1,2].

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2 Methods

2.1 Pendulum dynamics

We consider a pendulum with state

$$x_t = (\theta_t, \dot{\theta}_t),$$

where θ_t is the angle and $\dot{\theta}_t$ is the angular velocity. The action u_t is a bounded torque. The goal is to move the pendulum from the downward position to the upright position and stabilize it there. This is a standard nonlinear control problem frequently used as a benchmark in control and reinforcement learning [4,5].

We use known dynamics with discrete-time Euler integration:

$$\ddot{\theta}_t = \frac{3g}{2l}\sin(\theta_t) + \frac{3}{ml^2}u_t - b\dot{\theta}_t, \tag{1}$$

$$\dot{\theta}_{t+1} = \dot{\theta}_t + \Delta t\,\ddot{\theta}_t, \tag{2}$$

$$\theta_{t+1} = \theta_t + \Delta t\,\dot{\theta}_{t+1}. \tag{3}$$

After each step, the angle is wrapped to the interval $[-\pi, \pi]$. In our implementation, the upright target corresponds to $\theta = 0$ after wrapping.

2.2 Model Predictive Control

We use Model Predictive Control with horizon H [1–3]. At each time step, the controller samples candidate action sequences

$$\mathbf{u} = (u_0, u_1, \ldots, u_{H-1}),$$

rolls out the predicted trajectory under the dynamics model, computes its cost, and selects the sequence with the lowest cost. Only the first action is applied to the true system, after which planning is repeated from the new state. This receding-horizon strategy is the defining feature of MPC [1–3].

To optimize the action sequence, we use *random shooting*. Specifically, at each decision step, we sample N candidate sequences uniformly from the valid action range. This method is simple and computationally feasible for the pendulum because the state and action spaces are low-dimensional. Random-shooting and related sampling-based optimizers are widely used in model-based control, especially in low-dimensional settings and as baselines for more advanced planners [8–10].

2.3 Cost function

The controller minimizes the trajectory cost

$$J = \sum_{t=0}^{H-1}\left(q_\theta \tilde{\theta}_t^2 + q_\omega \dot{\theta}_t^2 + R u_t^2\right),$$

where $\tilde{\theta}_t$ is the wrapped angular error relative to the upright position, q_θ penalizes angle error, q_ω penalizes angular velocity, and R penalizes control effort. Quadratic stage costs of this form are standard in optimal control and MPC because they provide a simple way to balance state regulation against control effort [1–4].

In all experiments, we use $q_\theta = 1.0$ and $q_\omega = 0.1$, while varying the control penalty R.

2.4 Experimental setup

We study the effect of planning horizon and control penalty. The tested planning horizons are

$$H \in \{10, 20, 40\},$$

and the tested control penalties are

$$R \in \{0.001, 0.01, 0.1\}.$$

Each episode starts near the downward position and runs for 200 time steps. At each step, the controller samples 256 candidate action sequences and chooses the best one according to the model-predicted cost. The torque is bounded to $[-2, 2]$. We repeat each setting over 10 random seeds and report mean performance across runs.

The pendulum parameters are set to $g = 9.81$, $m = 1.0$, $l = 1.0$, damping $b = 0.05$, and time step $\Delta t = 0.05$.

2.5 Evaluation metrics

We report the following metrics:

- **Success rate:** fraction of trials in which the pendulum is near upright and nearly stationary during the final part of the episode.
- **Final angle error:** wrapped absolute angular error at the last time step.
- **Control effort:** cumulative squared torque, $\sum_t u_t^2$.
- **Swing-up time:** first time step at which the pendulum enters the target region.

A trial is considered successful if during the final 20 steps it satisfies

$$|\tilde{\theta}_t| < 0.2, \qquad |\dot{\theta}_t| < 0.5.$$

3 Results

3.1 Qualitative controller behavior

Figure 1 shows a representative trajectory under MPC. The controller initially applies alternating torques to build momentum, producing large oscillations in angle and angular velocity. After roughly 100 time steps, the pendulum reaches the upright region and remains stabilized near the target. The apparent discontinuities in the angle plot are caused by wrapping the angle to the interval $[-\pi, \pi]$, rather than by physical jumps in the state. This behavior is consistent with the non-myopic nature of swing-up control discussed in the optimal control literature [1,4].

3.2 Effect of planning horizon

Figure 2 shows success rate as a function of planning horizon for different control penalties. The effect of horizon is not purely monotonic. Increasing the horizon from 10 to 20 improves robustness substantially, especially when the control penalty is high. In particular, the setting $(H = 10, R = 0.1)$ fails completely, while all settings with $H = 20$ achieve a success rate of 1.0.

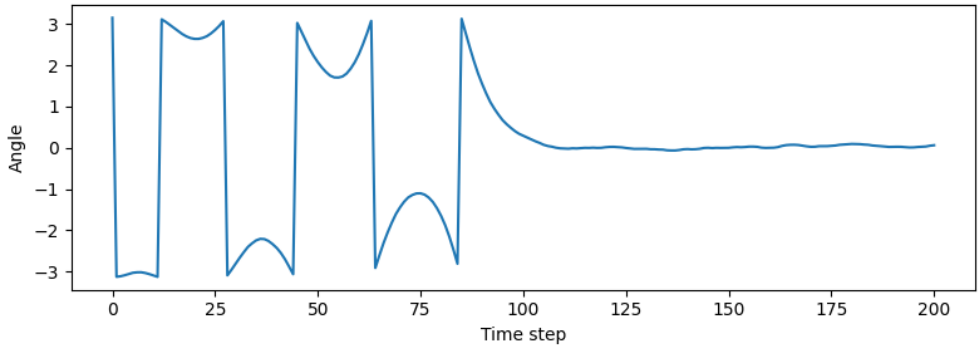
However, increasing the horizon further to 40 does not improve performance further in this experiment. For smaller control penalties, success rate slightly decreases at $H = 40$. This suggests that, for this task and random-shooting planner, an intermediate horizon gives the best trade-off between lookahead and optimization difficulty. This trade-off is broadly consistent with the MPC literature, where longer horizons can improve decision quality but also make planning harder under limited computational budget [1–3].

3.3 Effect of control penalty

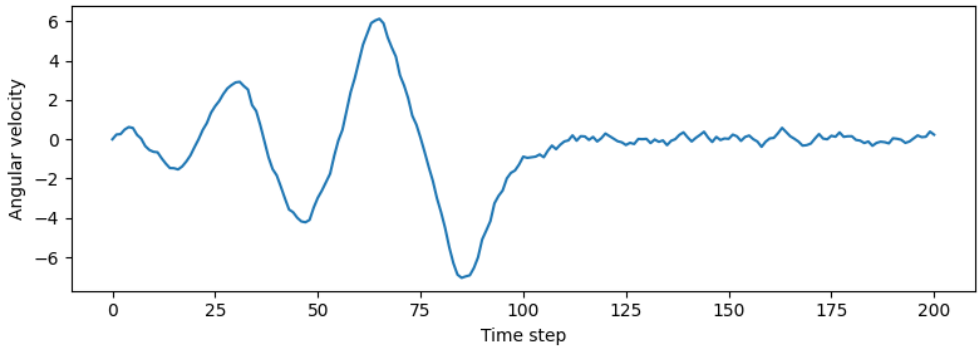
The control penalty R governs the trade-off between aggressive actuation and smooth control [1–3]. At horizon $H = 10$, a large penalty of $R = 0.1$ makes the controller too conservative and prevents successful swing-up. In contrast, at horizon $H = 20$, all three penalties achieve perfect success, with higher penalties reducing control effort. The lowest mean control effort is obtained at $(H = 20, R = 0.1)$.

At horizon $H = 40$, higher penalties again perform best, while smaller penalties show a slight drop in success rate. Overall, the results suggest that the best parameter choice is not simply the smallest penalty, but rather a combination of sufficient planning horizon and appropriately regularized control.

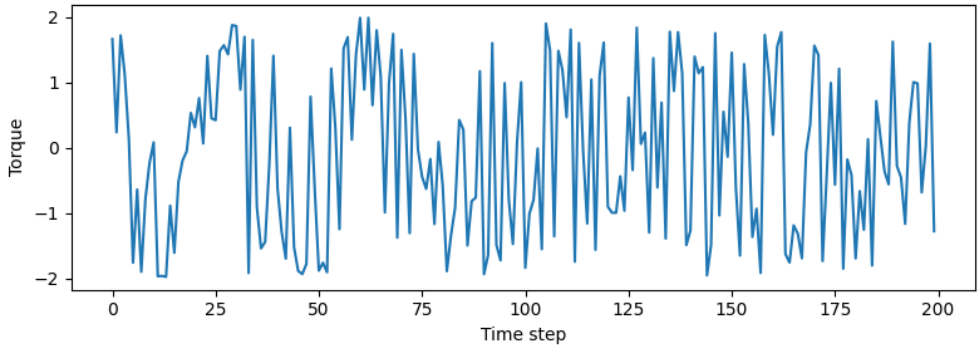
Table 1 summarizes the main quantitative results.



(a) Angle over time.



(b) Angular velocity over time.



(c) Torque over time.

Figure 1: Example MPC trajectory on the pendulum swing-up task. The controller first builds momentum, then brings the pendulum close to the upright state and stabilizes it.

3.4 Discussion of results

The experiments show that MPC is effective on pendulum swing-up, but performance depends strongly on both planning horizon and control penalty. The clearest pattern is that horizon $H = 20$ performs best overall: it is long enough to support non-myopic swing-up behavior, but not so long that random shooting becomes less reliable.

The control penalty has a similarly important effect. With too much penalty and too short a horizon, the controller becomes overly conservative. With a better horizon, however, a larger control penalty can reduce effort while preserving perfect success. These findings highlight that MPC performance

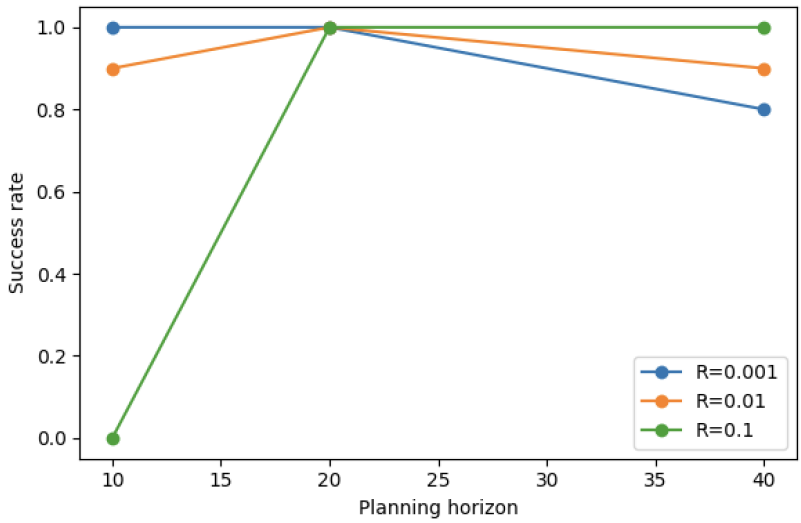


Figure 2: Success rate as a function of planning horizon for different control penalties. The best overall performance is obtained at horizon $H = 20$, where all tested penalties achieve perfect success.

Table 1: Summary of MPC performance across planning horizons and control penalties.

Horizon H	Penalty R	Success Rate	Final Angle Error	Control Effort
10	0.001	1.00	0.0051	300.05
10	0.01	0.90	0.0190	298.27
10	0.1	0.00	1.3301	397.71
20	0.001	1.00	0.0217	267.31
20	0.01	1.00	0.0268	250.67
20	0.1	1.00	0.0230	229.63
40	0.001	0.80	0.0454	297.24
40	0.01	0.90	0.0312	288.38
40	0.1	1.00	0.0267	291.50

depends not only on model accuracy, but also on the interaction between planning depth and cost design, which is a central theme in the MPC and optimal control literature [1–4].

4 Limitations

This project has several limitations. First, the dynamics model is assumed to be known exactly, which makes the problem easier than many realistic control settings where the model is learned or uncertain [5,8,10]. Second, the optimization method is random shooting, which is simple but not especially sample-efficient. More advanced planners such as the Cross-Entropy Method or gradient-based trajectory optimization could likely improve performance; related sampling-based and trajectory-optimization approaches have been studied extensively in model-based control and robotics [8–10]. Third, experiments are limited to a single environment, so conclusions should be interpreted as specific to pendulum swing-up rather than as general statements about all nonlinear control problems.

5 Conclusion

We investigated Model Predictive Control for pendulum swing-up using a known dynamics model and random-shooting optimization. The experiments were designed to study how planning horizon and control penalty affect controller performance. The main conclusion is that MPC can solve the task reliably when the planning horizon is long enough, while stronger control penalties reduce control

effort but can hinder swing-up success. Overall, the project shows that even a simple model-based controller can perform well on a nonlinear control problem when the model and cost are chosen appropriately, in line with standard results from MPC and optimal control [1–4].

References

[1] Eduardo F. Camacho and Carlos Bordons. *Model Predictive Control*. 2nd ed. London: Springer, 2007.

[2] James B. Rawlings, David Q. Mayne, and Moritz M. Diehl. *Model Predictive Control: Theory, Computation, and Design*. 2nd ed. Madison, WI: Nob Hill Publishing, 2017.

[3] Jan M. Maciejowski. *Predictive Control with Constraints*. Harlow, UK: Pearson Education, 2002.

[4] Dimitri P. Bertsekas. *Dynamic Programming and Optimal Control*. Vol. 1. 3rd ed. Belmont, MA: Athena Scientific, 2005.

[5] Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. 2nd ed. Cambridge, MA: MIT Press, 2018.

[6] Warren B. Powell. *Approximate Dynamic Programming: Solving the Curses of Dimensionality*. 2nd ed. Hoboken, NJ: Wiley, 2011.

[7] Dimitri P. Bertsekas and John N. Tsitsiklis. *Neuro-Dynamic Programming*. Belmont, MA: Athena Scientific, 1996.

[8] Grady Williams, Paul Drews, Brian Goldfain, James M. Rehg, and Evangelios A. Theodorou. "Information-Theoretic Model Predictive Control: Theory and Applications to Autonomous Driving." In *2017 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 1714–1721, 2017.

[9] Evangelos A. Theodorou, Jonas Buchli, and Stefan Schaal. "A Generalized Path Integral Control Approach to Reinforcement Learning." *Journal of Machine Learning Research*, vol. 11, pp. 3137–3181, 2010.

[10] Marc Peter Deisenroth, Gerhard Neumann, and Jan Peters. "A Survey on Policy Search for Robotics." *Foundations and Trends in Robotics*, vol. 2, no. 1–2, pp. 1–142, 2013.

[11] Boris Polyak and Anatoli Juditsky. "Acceleration of Stochastic Approximation by Averaging." *SIAM Journal on Control and Optimization*, vol. 30, no. 4, pp. 838–855, 1992.

[12] Harold J. Kushner and G. George Yin. *Stochastic Approximation and Recursive Algorithms and Applications*. 2nd ed. New York: Springer, 2003.

A Appendix / supplemental material

This appendix includes per-seed results(uploaded @github), implementation details, and exact commands used to reproduce the experiments.

A.1 Reproducibility details

The full experiment can be run from the project root with:

```
python -m venv .venv
pip install -r requirements.txt
python main.py
```

On Unix-based systems, the virtual environment can be activated with `source .venv/bin/activate`; on Windows, use `.venv\Scripts\activate`.

The project repository is available at `https://github.com/sheersh-ss/mpc_pendulum_project`

The code saves raw metrics to `results/raw/` and figures to `results/figures/`. The notebook `quick_analysis.ipynb` can be used to inspect results and export LaTeX-ready tables.

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

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   Justification: The abstract and introduction state that the paper studies MPC for pendulum swing-up and evaluates the effect of planning horizon and control penalty; these match the scope of the experiments.

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   Justification: The paper specifies the dynamics, cost function, horizons, penalties, episode length, number of sampled sequences, torque bounds, and number of seeds, and the appendix includes exact run commands.

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   Answer: [No]

   Justification: Results are averaged over multiple seeds, but the current draft does not yet include standard deviations or confidence intervals in the plots or tables.

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